

Cloud Computing

Cloud Computing Models

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- Cloud Computing Deployment Models
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud
 - Community Cloud
- Cloud Computing Service Models
 - Infrastructure as a Service (IaaS)
 - Platform as a Service (PaaS)
 - Software as a Service (SaaS)

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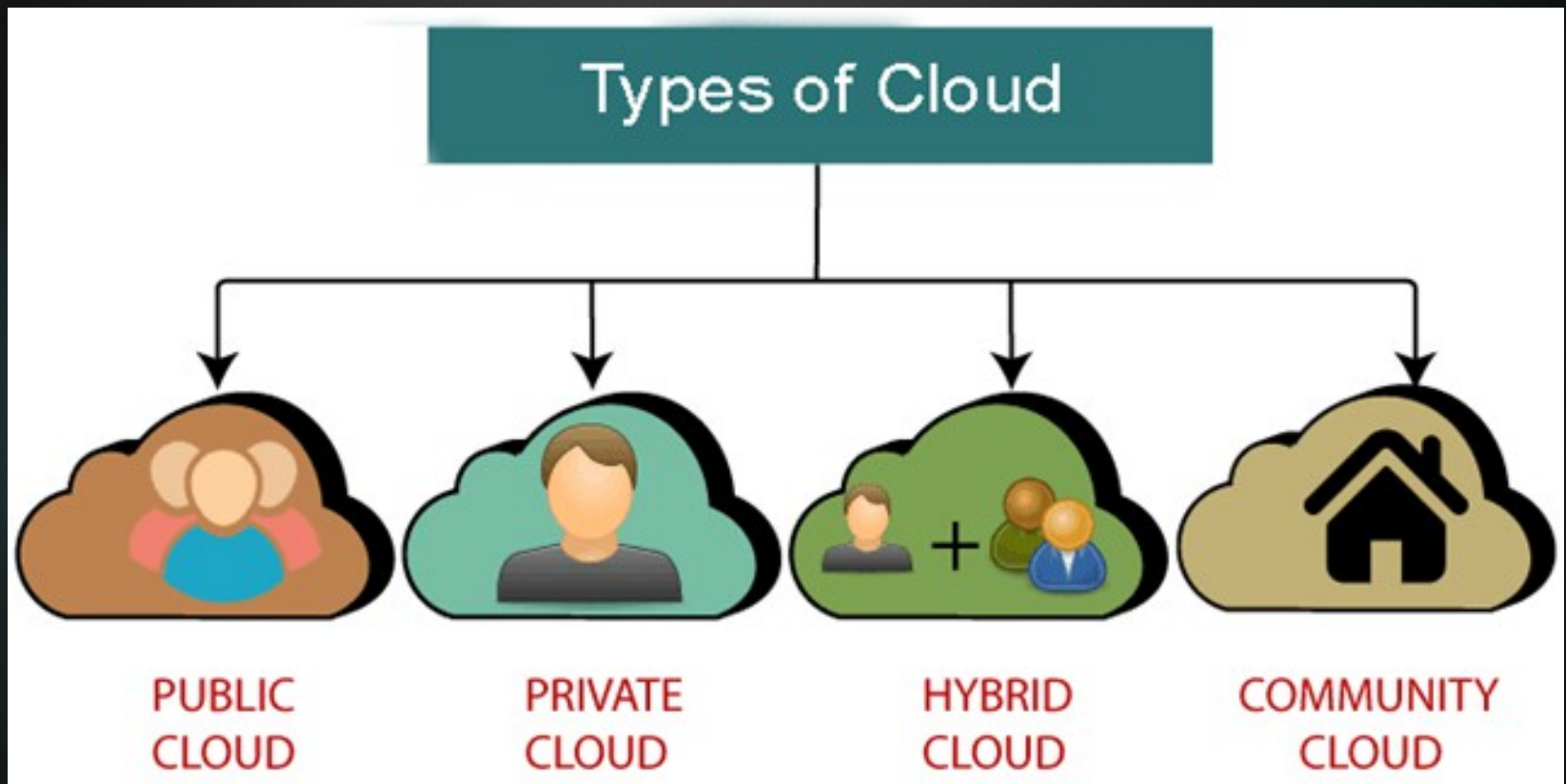
Cloud Computing Deployment Models

- Cloud computing is usually described in two ways.
 - Cloud Deployment Model
 - Cloud Service Model
- There are mainly four types of cloud computing deployment model
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud
 - Community Cloud

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Cloud Computing Deployment Models



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Cloud Computing Deployment Models: Public

- **Public Cloud** – It is a standard cloud computing model where resources provided by a service provider are shared across different clients.
- To share the computing resources the data of each tenant remains isolated from other tenants.
- The Public Cloud services are generally free for testing and after that, it works on a pay-as-you-go basis.

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Cloud Computing Deployment Models: Public

- For example: if you rent 10GB public cloud storage, you will get requested capacity of a disk from a bigger chunk of single storage device that is logically divided into small parts and distributed to multiple users isolated to make it private across different clients.
- In the public cloud, systems and services are accessible to the general public. For example, Google, Amazon, Microsoft etc.
- Client can use to deploy web-based office applications, storage, and many more applications.

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Cloud Computing Deployment Models: Public

- Advantage
 - Scalable: Resources can be increased and reduced on-demand
 - Reliable: High availability model, no single point of failure
 - Cost efficient: Public cloud is less expensive because it shares same resources with many customers.
- Disadvantage
 - Poor Customizable: It is comparatively less customizable than private cloud
 - Less Security: it cannot be used for private and sensitive information

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Cloud Computing Deployment Models: Private

- **Private Cloud** – Private Cloud has all resources dedicated to your application or organization.
- It gives complete control for data security and privacy. Private Cloud provides an additional layer of protection to process sensitive and confidential information.
- It consists of an underlying pool of a computing resource which will provide the computing power as a service within a virtualization environment.

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Cloud Computing Deployment Models: Private

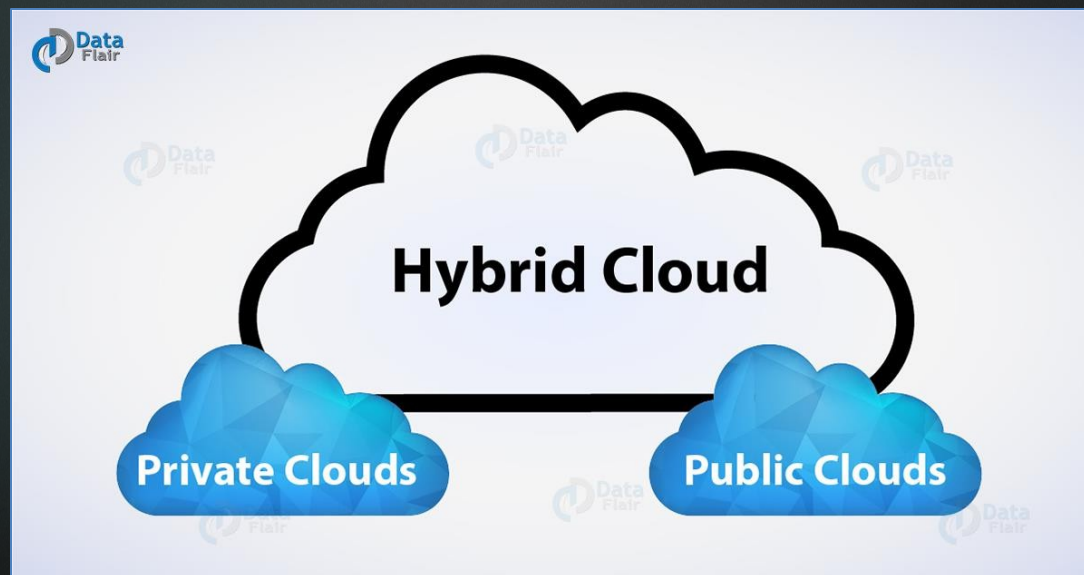
- Advantage
 - Better Security: Resources are dedicated and not shared with anyone
 - More Control: Private cloud has more control on its resources and hardware
 - Reliable: Good performance due to virtual partitions
- Disadvantage
 - Expensive: It provides secured and more features, so it's more expensive than a public cloud and required IT expertise to manage it.
 - Restriction: Private cloud is very difficult to deploy globally and it can be accessed locally only

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Cloud Computing Deployment Models: Hybrid

- **Hybrid Cloud** – It is a combination of Public Cloud and Private Cloud.
- Non-critical activities are performed using public cloud while the critical activities are performed using private cloud.



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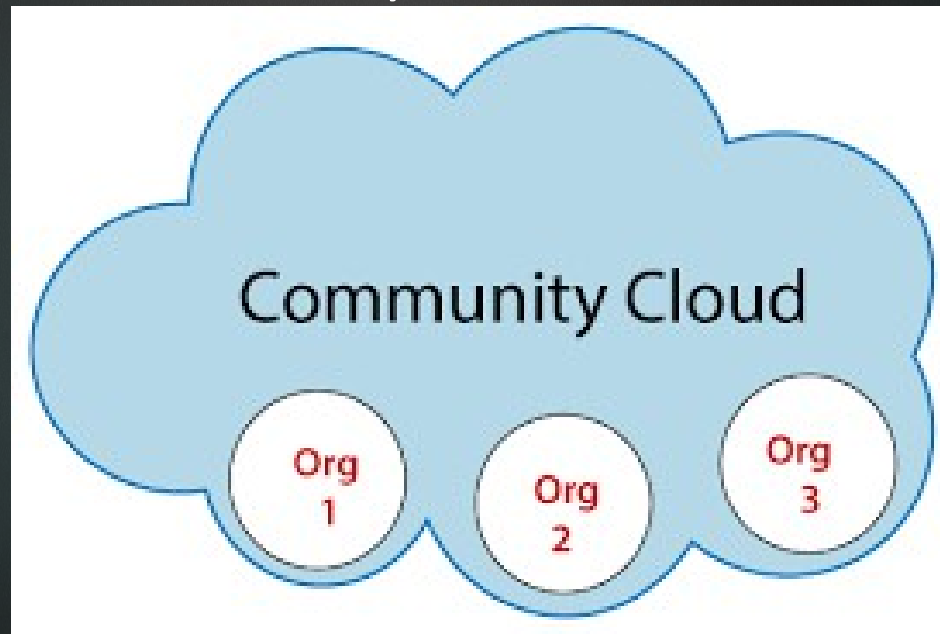
Cloud Computing Deployment Models: Hybrid

- Advantage
 - Scalable: You can increase your services starting from a Public cloud and Scale to Private as per demand.
 - Cost efficient: You don't have to setup 100% infrastructure for private cloud
 - Security: You can continue to run confidential processes in the Private Cloud.
- Disadvantage
 - Security: There are small chances of security breaching through public cloud.

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- ▶ Cloud Computing Deployment Models: Community
 - Community Cloud – It allows system and services to be accessible by group of organizations.
 - It shares the infrastructure between several organizations from a specific community.



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- ▶ Cloud Computing Deployment Models: Community
 - It is managed internally and operated by several organizations or by the third party or combination of them.
 - For joint business organizations, ventures, research organizations, and tenders, a community cloud is an appropriate solution.

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Cloud Computing Deployment Models: Community

- Advantage
 - Scalable: It is mostly compatible with every user and they can modify as per their use.
 - Availability: Community cloud gives an infrastructure to share cloud resources between several organizations.
- Disadvantage
 - Security: All data is located at one place, one must be careful in storing data in community cloud because it might be accessible to others.

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Cloud Computing Deployment Models:

